



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JULY 2022**

Course : EC3114 (Software Modelling and Analysis)

Time : 2.00 pm – 5.00 pm
(3 hours)

Lecturer : Noor Zuhaili bt Md Yasin

Date : 5 January 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 3 printed pages
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. List FOUR (4) purposes of requirement engineering analysis. (4 MARKS)
 - b. State any FIVE (5) sections commonly found in software requirement specifications (SRS). (5 MARKS)
 - c. Explain THREE (3) phases of requirement engineering. (6 MARKS)
 - d. Discuss the interview fact finding method in requirement engineering. Include the criteria during the interview and types of information uncovered in interview session. (10 MARKS)
2.
 - a. Define the concept of data dictionary in software modeling and analysis. (4 MARKS)
 - b. Explain data structures in data dictionary. (4 MARKS)
 - c. Discuss the following object-orientation concepts: (8 MARKS)
 - i. Encapsulation and Methods
 - ii. Classes.
 - d. Model a diagram to illustrate the difference between functional, structural, and behavioural models. Provide at least THREE (3) examples of types of diagrams for each model. (9 MARKS)

(For Question 3 and Question 4)

Case Study: XYZ International Gastronomy Festival Tickets Selling System.

3. XYZ International Gastronomy Festival is a famous foodie festival, held in every year. Since its first edition in 2017, it has gone through several major changes regarding its structure, length and location, but the tickets have always been sold in a traditional way: through two events agencies. The organizers decided to completely modernize the tickets selling system and created the following concept.

From this year on, the tickets will be sold in two distinct ways: traditionally, i.e. by the one event agency and in electronic format directly on the festival website. All parties will have access to the same unique tickets database of the new system, to avoid double selling. Moreover, the system will have to be extended to support not only in Arabic language, but also in English, French and Italian. Since tickets will also be sold online, SecurePayment Inc. will be contracted to provide and ensure the security of the online payment service. The XYZ event manager will take care and negotiate all these details with the involved parties.

Additionally, upon arrival at the festival venue, each participant has to self-check in at a touch screen terminal, which scans the barcodes on his/her ticket and issues a bracelet with an electronic chip. This can be used to load money, such that whenever (s)he wants to purchase snacks or beverages, (s)he does not have to use cash anymore, thus reducing waiting times. This measure was initiated by the program manager and will be deployed by WristSolutions Inc. Lastly, according to the cantonal laws, the way the payment transactions are performed has to be audited by an external company at the end of the festival, since this is a public event.

- a. List EIGHT (8) stakeholders of the new XYZ International Gastronomy Festival Tickets Selling System. (8 MARKS)
- b. Draw a context diagram of the new XYZ International Gastronomy Festival Tickets Selling System. Make sure you label the actors, the relationships between them as well as their relationships to the system. (10 MARKS)
- c. According to the case study, identify:
 - i. ONE (1) goal. (1 MARK)
 - ii. TWO (2) functional requirements. (3 MARKS)
 - iii. TWO (2) non-functional requirements. (3 MARKS)

4. a. List any SIX (6) methods how you can gather further requirements that help you build the XYZ International Gastronomy Festival new tickets selling system. (6 MARKS)
- b. Refer to the methods that you have listed in Q4(a), choose any FOUR (4) and state: (12 MARKS)
- i. the reason for suggesting the method,
 - ii. who are the stakeholders, and
 - iii. how you would implement the method in practice for XYZ International Gastronomy Festival.
- c. Identify all the use cases in the XYZ International Gastronomy Festival case study and represent them in a UML use case diagram. (7 MARKS)

-END OF QUESTION PAPER-